

METAL-CLAD MV SWITCHGEAR · KYN28A PLATFORM

PLGEAR

Air-insulated withdrawable switchgear with vacuum circuit breakers for primary distribution

12 & 24 kV · up to 3,150 A · up to 31.5 kA



31.5 kA

SHORT-TIME
WITHSTAND · 3 S

LSC2B

SERVICE CONTINUITY ·
PM

IAC AFLR

INTERNAL ARC
CLASSIFIED

VCB

ABB VACUUM
BREAKERS

01 · COMPANY PROFILE

ENGINEERING THE FUTURE OF POWER WITH PRECISION AND INNOVATION

Since our inception in 2012, Powerline has been driven by a singular mission: to serve as the reliable backbone of the regional and Egyptian power sector. We are not merely a contracting and supply firm — we are a fully integrated engineering partner specialising in the design and manufacture of Medium Voltage (MV) and Low Voltage (LV) power distribution systems, adhering to the most stringent international technical standards.

A Legacy of Reliability

Built on foundations of commitment and precision, Powerline has established itself as a hallmark of quality in major national and industrial projects. Our track record reflects a consistent ability to overcome complex engineering challenges and deliver customised solutions for the modern electrical grid.

Solutions Beyond Expectations

Our MV switchgear, Ring Main Units, compact substations and LV distribution boards are characterised by exceptional operational efficiency and a long life cycle. We bridge mechanical durability with digital intelligence, ready for integration with the latest smart-grid and control technologies.

Quality is not a department — it is our work culture



Production & Manufacturing

Advanced production lines ensure the highest accuracy in the assembly and calibration of MV & LV switchgear, in full compliance with IEC international standards.



Engineering & Design

Our design engineers develop layouts that prioritise maximum safety, operational continuity and seamless serviceability across the installation's life.



Testing & Quality Control

Every unit leaving our facilities undergoes rigorous routine testing to ensure durability under the harshest operating conditions — total peace of mind for our clients' investments.

2012

ESTABLISHED · CAIRO

2MANUFACTURING
FACILITIES**2,200+**

PROJECTS DELIVERED

OEMMV SWITCHGEAR · ABB-
CERTIFIED LV

“ Powerline: Performance Excellence, Execution Precision & Sustainable Solutions. ”

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How to use this catalogue

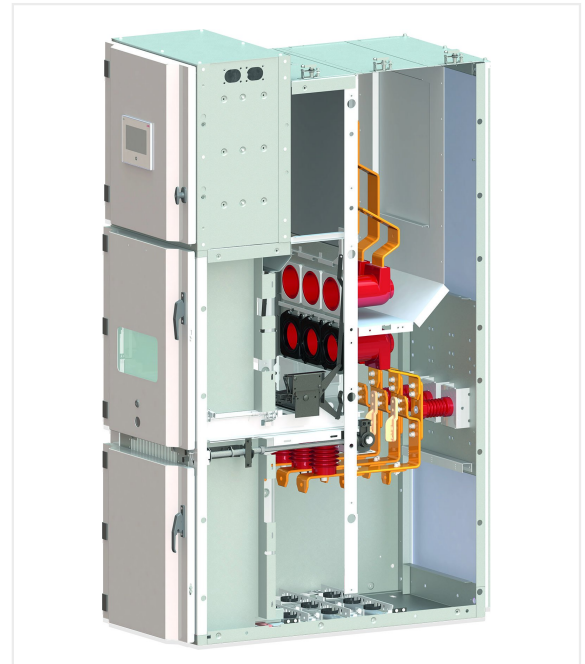
Sections 06–09 cover the panel and its ratings.
Sections 11–13 detail the apparatus installed inside.
Sections 14–16 support unit selection and installation planning.

02 · PRODUCT OVERVIEW

PLGEAR AT A GLANCE

PLGEAR is Powerline's metal-clad, air-insulated switchgear for primary distribution at 12 and 24 kV — built on the globally proven KYN28A platform and equipped with ABB apparatus throughout: VD4 vacuum circuit breakers, Relion® protection relays and ABB instrument transformers.

- Metal-clad, withdrawable design — each function in its own earthed steel compartment
- Withdrawable vacuum circuit breaker truck: service, test and racked-out positions
- Loss of service continuity class **LSC2B** with metallic partitions (PM)
- Internal arc classification **IAC AFLR** — all four sides classified
- All installation, operation and maintenance from the front — back-to-wall installation
- Earthing switch with full short-circuit making capacity
- Complete safety interlocking between breaker, truck, earthing switch and doors



PLGEAR panel — compartmented, metal-clad construction

ABB inside

The same apparatus family used in ABB UniGear ZS1 — assembled, wired and routine-tested by Powerline in Egypt.

24 kV

RATED VOLTAGE

3,150 ABUSBAR CURRENT UP
TO**31.5 kA**

SHORT-TIME · 3 S

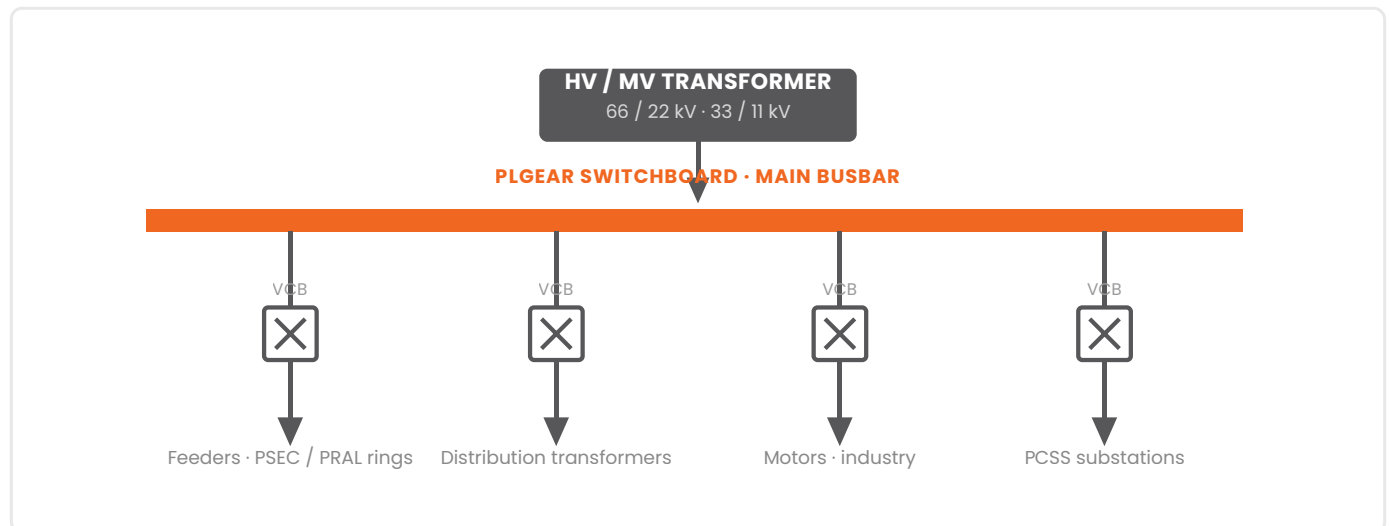
80 kAPEAK WITHSTAND UP
TO

Where it sits in the family. PLGEAR is Powerline's primary-distribution switchgear — upstream of the PSEC / PRAL ring main units and PCSS compact substations that serve the secondary network. One supplier, one engineering standard, from primary substation to final feeder.

03 · APPLICATIONS

FIELD OF APPLICATION

PLGEAR receives, protects and distributes medium voltage power at the heart of the network — the primary substation. Vacuum circuit breakers on every way protect cables, overhead lines, transformers, motors, generators and capacitor banks.



Typical primary substation — incomer from the HV/MV transformer, breaker-protected outgoing feeders

Typical duties

- Incoming and outgoing feeder protection with vacuum circuit breakers
- Bus-tie and riser arrangements for sectionalised busbar schemes
- Transformer, motor, generator and capacitor-bank protection
- Utility metering with dedicated measurement units

Typical environments

Utility primary substations	Industrial plants
Power generation	Water & infrastructure
Commercial complexes	Oil & gas facilities
Data centres	

SAFETY BY DESIGN

Metal-clad construction means every energised function lives behind earthed steel. Add class-leading internal arc performance and a complete interlocking scheme, and PLGEAR protects the operator in every state of service.



Internal arc — IAC AFLR

Internal arc classified on all four sides (front, lateral, rear) up to 31.5 kA / 1 s, with optional gas exhaust duct for controlled venting.



LSC2B · PM segregation

Metallic partitions between busbar, breaker and cable compartments — one compartment can be opened while the busbar and adjacent panels stay in service.



Complete interlocking

Truck, breaker, earthing switch and doors are mechanically interlocked — the safe sequence is enforced by steel, not procedure.



Automatic shutters

Metallic shutters close over the live spouts automatically as the breaker truck is racked out — no access to live contacts, ever.



Making-capacity earthing

The cable earthing switch has full short-circuit making capacity; its position is shown by a mechanically coupled indicator.



Earthing busbar

An electrolytic-copper earthing busbar runs the full length of the switchboard for maximum personnel and installation safety.

IEC 62271-200

Designed, built and routine-tested to the standard for AC metal-enclosed switchgear up to 52 kV — with apparatus type-tested by ABB to IEC 62271-100 and related standards.

PRODUCT FEATURES



Technology

- Metal-clad, air-insulated, withdrawable design
- ABB VD4 vacuum circuit breakers
- Copper busbars, insulated with shrink-on sleeves at 24 kV
- Single-pole epoxy insulating bushings with automatic shutters
- Relion® digital protection relays



Durability

- Vacuum interruption — no arc chopping, no overvoltages
- Sturdy embedded-pole breaker construction
- Metallic compartment partitions (PM)
- Powder-coated sheet-steel enclosure, RAL 7035
- Type-tested apparatus throughout



Safety

- IAC AFLR internal arc classification
- Interlocked truck, breaker, earthing switch and doors
- Automatic shutters over live spouts
- Earthing switch with making capacity
- Full-length earthing busbar



Serviceability

- All operations and maintenance from the front
- Back-to-wall installation — no rear access needed
- Withdrawable breaker exchanged in minutes
- Cables accessible from the front, up to 12 per phase
- Local manufacture: fast delivery and lifetime support

Withdrawable by design

The circuit breaker sits on a truck with three defined positions — **service**, **test** and **racked-out**. Protection testing without touching the primary circuit; breaker exchange without de-energising the busbar.

3

 TRUCK
POSITIONS

100%

 FRONT
OPERATED

0

 REAR ACCESS
NEEDED

CONSTRUCTION & COMPARTMENTS

Each PLGEAR unit consists of three power compartments — circuit breaker, busbars and cables — plus a low voltage compartment for the secondary equipment. All compartments are segregated by metallic partitions and are accessible from the front.

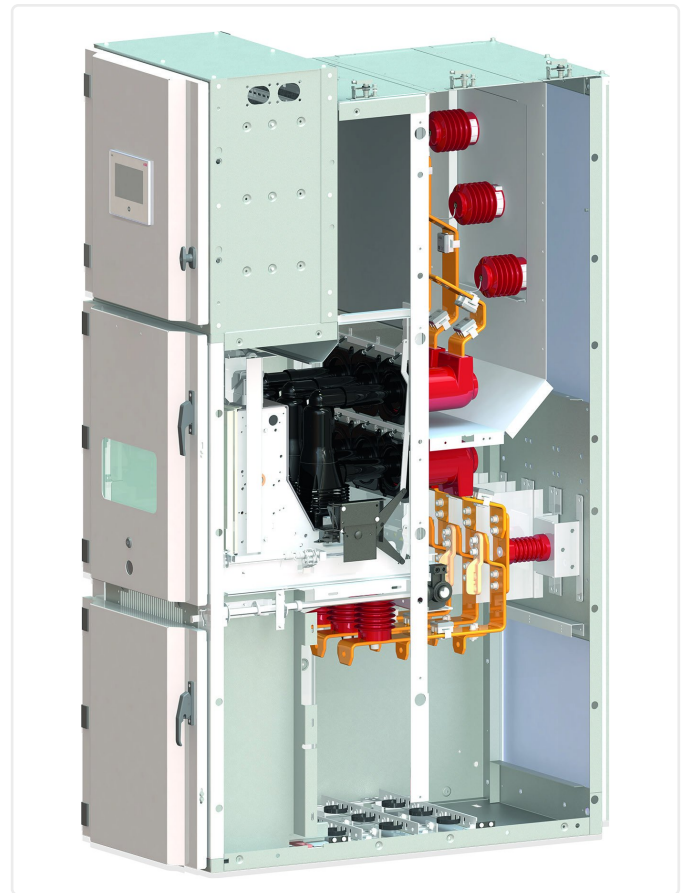
Compartment	Description
A • Busbar	Flat electrolytic-copper main busbars laid panel to panel, insulated with shrink-on sleeves; bolted joints under insulating covers
B • Circuit breaker	Withdrawable VD4 truck engaging single-pole epoxy bushings; automatic metallic shutters isolate the spouts when racked out
C • Cable	Front-access connection for single or three-core cables (up to 12 per phase); earthing switch with making capacity; optional CTs and surge arresters
D • Low voltage	Relion® protection relay, instruments, terminal blocks and auxiliary wiring, segregated from all power compartments

Arc gas management

Arc-proof panels are provided with a duct that evacuates the gases produced by an internal arc safely above or away from the operator area — several duct arrangements are available.

Earthing

A copper earthing busbar runs longitudinally through the switchboard; the enclosure, partitions and apparatus frames are all bonded to it.



Compartmented construction — breaker, busbar and cable zones behind metallic partitions

MAIN CHARACTERISTICS

Ratings of the PLGEAR switchboard at both voltage classes. Values apply to the standard indoor service conditions on page 10.

Characteristic	Unit	PLGEAR 12	PLGEAR 24
Rated voltage / rated insulation voltage	kV	12	24
Service voltage	kV	11	22
Rated power-frequency withstand voltage (1 min)	kV	28	50
Rated lightning impulse withstand voltage (1.2/50 µs)	kV	75	125
Rated frequency	Hz	50	50 / 60
Rated short-time withstand current (3 s)	kA	31.5	up to 31.5
Rated peak withstand current	kA	80	up to 80
Internal arc withstand current (1 s)	kA	up to 31.5	up to 31.5
Main busbar rated current	A	up to 3,150 ⁽¹⁾	up to 3,150 ⁽¹⁾
Circuit breaker rated currents	A	630 ... 2,500 ⁽²⁾	630 ... 2,500 ⁽²⁾
Loss of service continuity class	—	LSC2B	LSC2B
Partition class	—	PM (metallic)	PM (metallic)
Internal arc classification	—	IAC AFLR	IAC AFLR
Degree of protection — enclosure	—	IP43	IP4X
Degree of protection — partitions	—	IP2X	IP2X
Mechanical impact protection	—	IK10	IK10
Applicable standard	—	IEC 62271-200	IEC 62271-200
Installation	—	Indoor	Indoor
Enclosure colour	—	RAL 7035	RAL 7035

(1) 4,000 A on request. (2) 3,150 A available; higher ratings with forced ventilation on request. The VD4 vacuum circuit breaker covers the full current and voltage range, so the apparatus, partitioning and arc-classification data are common to both voltage classes. 12 kV values are those marked on the PLGEAR-12KV rating plate; 24 kV values follow the reference apparatus data and are confirmed per project configuration.

Vacuum interruption

Current interruption without arc chopping and without overvoltages — dielectric recovery is extremely rapid.

Insulated busbars

Busbars and branch conductors carry shrink-on sleeve insulation with covered joints at both voltage classes.

Panel widths

12 kV: 800 mm. 24 kV: 800 mm up to 2,000 A, 1,000 mm up to 3,150 A — see Dimensions, page 20.

STANDARDS & SERVICE CONDITIONS

PLGEAR is designed, manufactured and routine-tested in accordance with the IEC 62271 series; every installed apparatus is individually type-tested by its manufacturer.

Standard	Scope
IEC 62271-1	High-voltage switchgear and controlgear — Part 1: Common specifications
IEC 62271-200	Part 200: AC metal-enclosed switchgear and controlgear up to 52 kV — incl. LSC / PM classification and internal arc
IEC 62271-100	Part 100: Alternating-current circuit-breakers (VD4)
IEC 62271-102	Part 102: Disconnectors and earthing switches
IEC 61869	Instrument transformers — CTs, VTs and sensors
IEC 60529	Degrees of protection provided by enclosures (IP)

Service conditions

Condition	Specification
Installation	Indoor
Ambient temperature	-5 °C to +40 °C
Installation altitude	Up to 1,000 m (higher on request)
Relative humidity	Max. 95 %
Enclosure colour	RAL 7035
Factory acceptance test	Powerline standard FAT (one day)

Why it matters

- **Metal-clad per IEC 62271-200** — LSC2B-PM is the highest service-continuity class for air-insulated switchgear.
- **Type-tested apparatus** — breaker, relays and instrument transformers each carry their own type-test evidence.
- **Routine tests on every switchboard** — dielectric, mechanical, interlock and wiring verification before dispatch.

The electrical characteristics can vary for ambient conditions other than standard or when a higher degree of protection is specified — Powerline engineering confirms ratings per project.

INTERLOCKS & OPERATION

Every operating sequence is enforced mechanically. The truck, circuit breaker, earthing switch and doors permit only the safe order of operations.

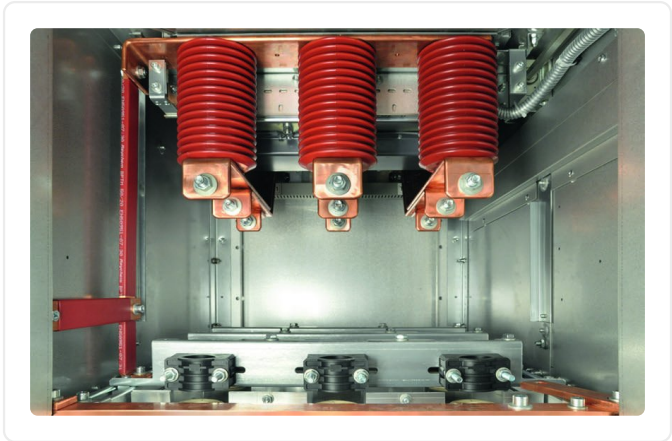
#	Interlock condition
1	The breaker truck can be racked in or out only with the circuit breaker open
2	The circuit breaker can close only with the truck fully in the service or test position
3	The earthing switch can close only with the truck racked out to the test/disconnected position
4	The cable compartment door opens only with the earthing switch in the ON position
5	The earthing switch cannot open while the cable compartment door is open
6	Metallic shutters close automatically as the truck moves from service towards racked-out

Locks & supervision

Key locks and padlocking facilities on truck, shutters and earthing switch support switching programmes; auxiliary contacts report every position to the control room.

Truck positions

Position	State
Service	Primary contacts engaged – breaker in the power circuit, ready for duty
Test	Primary contacts separated, secondary connected – full function testing without touching the network
Racked out	Truck withdrawn, shutters closed over live spouts – breaker removable for service



Single-pole epoxy bushings behind the automatic shutters

Inside PLGEAR

We use the same apparatus as ABB's UniGear ZS1 platform — the world's reference in air-insulated primary switchgear — assembled, wired and routine-tested by Powerline in Egypt.

VD4 vacuum circuit breaker · pages 13–14

Embedded-pole vacuum interruption, 630 – 2,500 A

Relion® protection & control · page 15

REF-series feeder relays and REX640 all-in-one protection

Instrument transformers & sensors · page 16

Block-type CTs, VTs and current/voltage sensors



ABB VD4 WITHDRAWABLE VACUUM CIRCUIT
BREAKER

VCB

VACUUM INTERRUPTION

2,500 A

BREAKER RATINGS UP TO

Relion®

DIGITAL PROTECTION

LPIT

SENSOR-READY

VD4 VACUUM CIRCUIT BREAKER

The ABB VD4 interrupts current in vacuum — used as both the breaking and the insulating medium. Interruption takes place without arc chopping and without overvoltages, and the dielectric strength recovers extremely rapidly.

Embedded poles

- Vacuum interrupters embedded in resin poles — sturdy, and protected against shocks, dust and humidity
- Ceramic-housed interrupter with copper-chrome contacts and metallic bellows — vacuum better than 10^{-5} Pa
- Radial and axial magnetic-flow techniques distribute the arc roots evenly over the contact surface

Protection duties

Cables, overhead lines, motors, transformers, generators and capacitor banks — one breaker family for every way in the switchboard, with a stored-energy spring mechanism for O-C-O duty cycles.

Withdrawable truck

The breaker is mounted on a truck with service / test / racked-out positions, interlocked with the panel — exchangeable in minutes without touching the busbar.



VD4 withdrawable vacuum circuit breaker

24 kV

RATED VOLTAGE

31.5 kA

BREAKING UP TO

VD4 RATINGS IN PLGEAR

Characteristic	Unit	VD4 — 24 kV class
Rated voltage	kV	24
Rated power-frequency withstand voltage	kV	50
Rated lightning impulse withstand voltage	kV	125
Rated frequency	Hz	50 / 60
Rated currents	A	630 · 1,250 · 1,600 · 2,000 · 2,500
Rated short-circuit breaking current	kA	up to 31.5
Interrupting medium	—	Vacuum
Pole construction	—	Embedded poles, epoxy resin
Operating mechanism	—	Stored-energy spring, motor-charged
Reference standard	—	IEC 62271-100

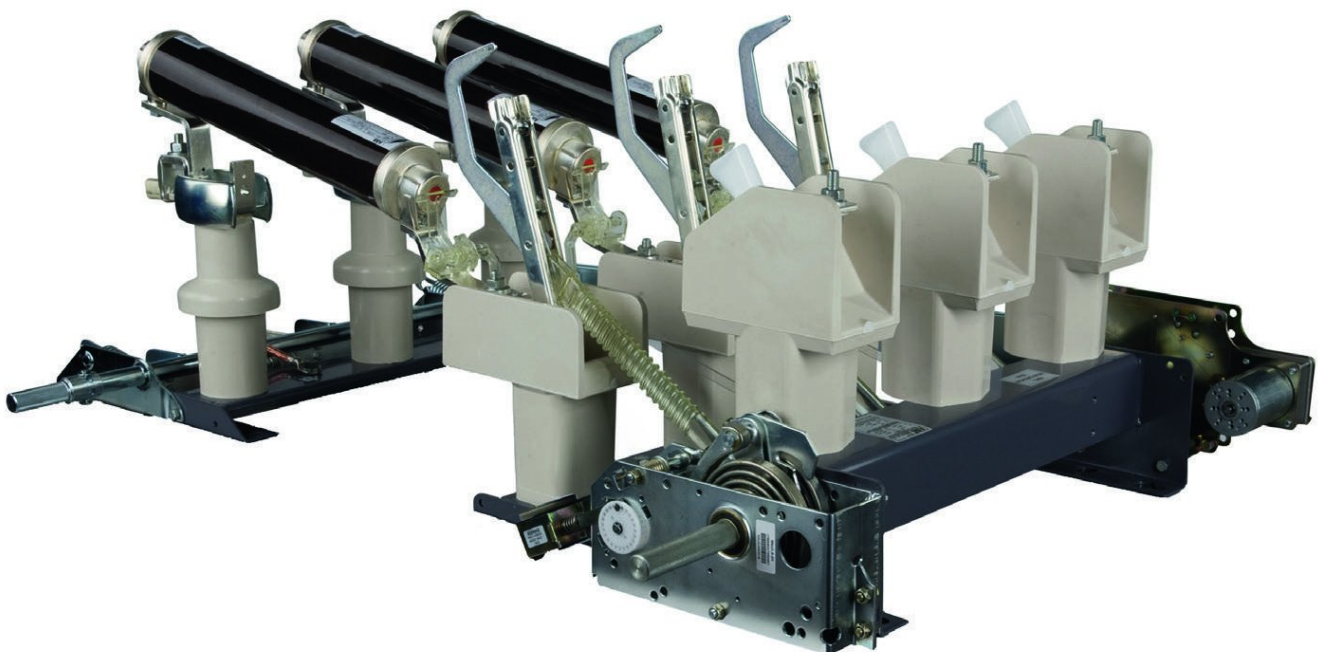
Ratings per the manufacturer's type tests; project-specific breaker selection is confirmed by Powerline engineering per feeder duty. Complete VD4 data on request.

Why vacuum

No gas handling, no arc products, minimal maintenance — and interruption behaviour that is gentle on cables, motors and transformers.

Alternatives

Contactors and switch-disconnector trucks are available for motor and auxiliary duties within the same panel system — ask Powerline engineering.



Withdrawable-truck isolating system with busbar connections

VD4 OPTIONAL ACCESSORIES

The VD4 operating mechanism accepts a range of factory-fitted electrical and mechanical accessories for remote control, supervision and key-managed safety. The same accessories apply across the current and voltage range, so a 12 kV and a 24 kV panel are specified and operated the same way. Specify the required accessories with the order — some cannot be retrofitted on site.



-MBO1 / -MBO2

Shunt opening release

Opens the breaker by remote command. A second, independent release can be fitted on a separate supply for duplicated tripping.



-MBC

Shunt closing release

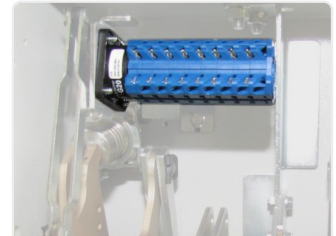
Closes the breaker by remote command. Permanently supplied, it provides the electrical anti-pumping function.



-MBU

Undervoltage release

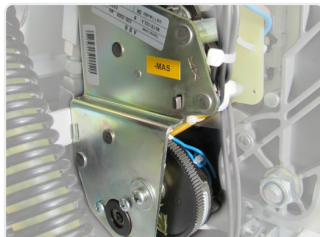
Trips the breaker when the supervised voltage disappears. Available instantaneous or with a time delay.



-BGB1

Auxiliary contacts

Open / closed signalling to the control system. Ten contacts as standard, up to twenty-four on the fixed version.



-MAS

Motor operator

Automatically recharges the closing springs after each operation, so the breaker is ready for the next close without manual charging.



KEY LOCK

Key lock in open position

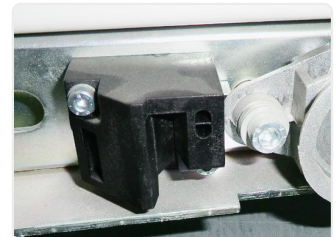
Locks the breaker open and releases the key only in that state — the basis of a key-interlock scheme across several panels.



-RLE2

Locking magnet on the truck

Electrically prevents the truck being racked in or out unless the interlocking conditions are satisfied.



DOOR INTERLOCK

Mechanical interlock with door

Prevents the compartment door being opened with the truck in service, and prevents racking in with the door open.

Also available

Opening solenoid (-MBO3) for self-supplied protection relays · transient and position contacts (-BGB4, -BGT3) · opening and closing pushbutton protection · locking magnet on the operating mechanism · interlock for fixed circuit breakers · motor-driven truck (-MAT) · STU shunt test unit for coil supervision.

Control supplies

Releases are available for 24, 30, 48, 60, 110...132 and 220...250 V DC, and 48, 60, 110...127 and 220...250 V AC at 50–60 Hz, operating between 65 % and 120 % of rated voltage. State the auxiliary voltage on the enquiry.

Accessory data per the VD4 apparatus documentation; availability and combination depend on the operating mechanism and the panel configuration, and are confirmed at order stage.

12 · PROTECTION & CONTROL

RELION® PROTECTION & CONTROL

Each PLGEAR unit carries an ABB Relion® digital relay in its low voltage compartment — protection, control, measurement and supervision in one device.

- **REF-series feeder relays** for overcurrent, earth-fault, voltage and frequency protection of incomers and feeders
- **REX640** for demanding applications — fully modular, all-in-one protection and control
- Transformer, motor and capacitor protection variants per unit duty
- IEC 61850 communication for station automation and SCADA integration
- Disturbance recording, event logs and condition monitoring built in

Configured and tested by Powerline

Relay selection, setting files and secondary wiring are engineered per project, functionally tested during FAT, and documented in the delivery dossier.



Relion® REF-series feeder protection



REX640 — modular all-in-one protection

INSTRUMENT TRANSFORMERS & SENSORS

Measurement feeds protection. PLGEAR accepts the full range of block-type current transformers, voltage transformers and modern low-power sensors.

- Block-type CTs to DIN dimensions, metering and protection cores per project
- Single-pole VTs, fixed or withdrawable, with protection fuses
- **Low-power instrument transformers (LPIT)** – compact current/voltage sensors with wide dynamic range, one sensor covering many ratings
- Ratios, burdens and accuracy classes engineered to the utility metering code and protection scheme

Powerline cast-resin alternative

Powerline's own P-LZZBJ9 current transformers and P-JDZX voltage transformers (IEC 61869) are available for metering duties – see the Powerline Instrument Transformers catalogue.



Block-type CTs, VTs and sensor family

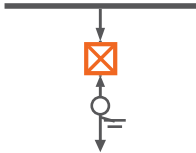


Combined current/voltage sensor (LPIT)

14 · TYPICAL UNITS

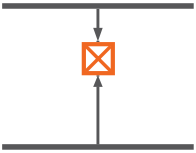
TYPICAL UNITS

Standard functional units combine into any single-busbar scheme. All units share the same compartment layout, interlocking and front-access operation.



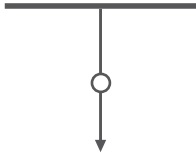
IF

Incoming / outgoing feeder. Withdrawable VD4 breaker with CTs, earthing switch and cable connection – the workhorse of the switchboard.



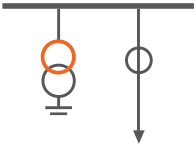
BT

Bus-tie. Couples two busbar sections through a withdrawable breaker for sectionalised operation.



R

Riser. Busbar elevation unit completing the bus-tie arrangement; with or without measurement.



M

Measurement. Withdrawable or fixed VTs and metering CTs for busbar measurement and revenue metering.

Unit availability – 24 kV

Unit	800 mm panel	1,000 mm panel
IF – incoming / outgoing feeder	630 – 2,000 A	630 – 2,500 A (3,150 A)
BT – bus-tie	630 – 2,000 A	up to 2,500 A
R / RM – riser (with measurement)	630 – 2,000 A	up to 3,150 A
M – measurement	Available in both widths	
IFD / IFDM – direct feeder (with measurement)	Selected ratings – per project	

Availability per the reference platform tables; the exact unit line-up is engineered per project single-line diagram.

SMART SOLUTIONS

PLGEAR is ready for the digital substation: sensors instead of iron-core transformers, IEC 61850 communication, and monitoring that turns maintenance from scheduled to condition-based.

- **Current & voltage sensors (LPIT)** — linear response over a wide range; fewer variants, faster engineering
- **IEC 61850 station bus** — relays share signals and interlocking over the network, GOOSE-fast
- **Condition monitoring** — temperature and diagnostic data from the switchboard reduces outages and site visits
- **Remote control** — breaker and earthing-switch supervision integrated with SCADA / RTU



Combined sensor with standardised RJ-45 interface

Scaled to the project

From simple local metering to a fully digital bay — Powerline configures the protection, measurement and communication level your network operator requires.

61850

IEC STATION BUS

GOOSE

PEER-TO-PEER
INTERLOCKING

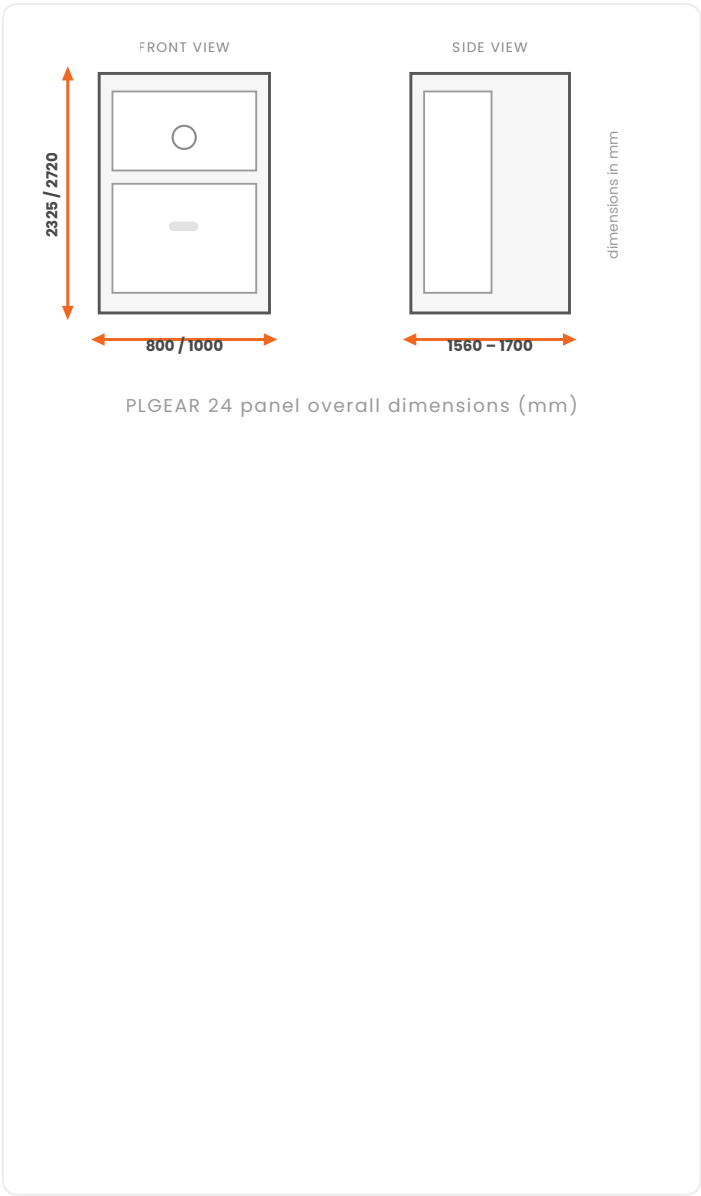
LPIT

SENSOR TECHNOLOGY

24/7

REMOTE SUPERVISION

PANEL DIMENSIONS



Dimension	PLGEAR 12	PLGEAR 24
Width — up to 2,000 A	800 mm	800 mm
Width — up to 2,500 / 3,150 A	1,000 mm	1,000 mm
Height — LV compartment 705 mm	2,230 mm	2,325 mm
Height — LV compartment 1,100 mm	—	2,720 mm
Height with gas exhaust duct	—	2,733 mm
Depth	1,500 mm	1,560 – 1,700 mm

12 kV values are the standard PLGEAR-12KV panel: 800 mm wide × 1,500 mm deep × 2,230 mm high. Entries marked — are configuration options offered on the 24 kV panel; contact Powerline engineering for the equivalent 12 kV arrangement.

Back-to-wall installation: all operation, cable termination and maintenance from the front. Cables — single or three-core, up to twelve per phase depending on cross-section.

Room planning: allow for the gas exhaust duct arrangement, floor openings per the cable layout, and the front operating aisle — layout drawings are issued with every project.

INSTALLATION & COMMISSIONING



- **Factory-assembled shipping sections** keep site work to coupling, connection and testing
- **No rear access required** — the switchboard installs against the wall
- **Busbars laid panel-to-panel** without special connection clamps
- **Front cable termination** with generous compartment space and optional cable supports
- **Documentation:** as-built drawings, single-line diagram, relay settings and test records with every delivery

Service & spares

Withdrawable apparatus makes service fast: a spare breaker truck covers the whole switchboard. Powerline supports commissioning, training and lifetime maintenance from Cairo.

Service conditions: indoor · $-5...+40\text{ }^{\circ}\text{C}$ · altitude up to 1,000 m (higher on request) · humidity max. 95 %.

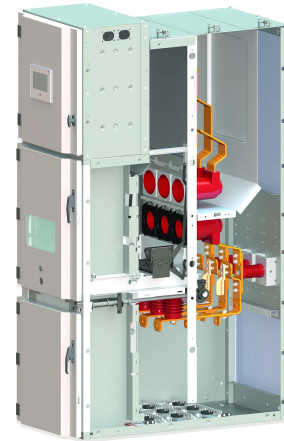
QUALITY & TESTING

Routine tests — every switchboard

- Visual inspection, dimensional and build verification
- Power-frequency voltage test on the main circuit
- Voltage test on auxiliary and control circuits
- Mechanical operation of trucks, shutters, earthing switches and interlocks
- Secondary injection and functional test of protection relays
- Documented factory acceptance test (FAT) — open to client witnessing

Type-tested foundations

- Panel system designed per IEC 62271-200 (LSC2B · PM · IAC AFLR)
- VD4 breakers, Relion® relays and instrument transformers individually type-tested by ABB



Every function verified before dispatch

Documentation with every delivery

Routine test certificates, as-built drawings, single-line diagram, relay setting files, operating manual and apparatus type-test evidence.



Performance excellence. Execution precision. Sustainable solutions.

From engineering and manufacture to testing, delivery and lifetime support — Powerline is your integrated partner for MV & LV power distribution in Egypt and the region.

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For technical enquiries or quotations, please get in touch with our team — we respond with engineered solutions, not just prices.

24 kV

RATED VOLTAGE

3,150 A

BUSBAR UP TO

31.5 kA

SHORT-TIME · 3 S

IAC AFLR

INTERNAL ARC
CLASSIFIED